csd5748

ΗΥ-119 - Γραμμική Άλγεβρα - Άσκηση 1.

· **1η Άσκηση:**

Έχουμε: $\left\{\begin{array}{l} 2u-3v+0w=3 \\ 4u+5v+1w=7 \\ 2u-1v-3w=5 \end{array}\right. \overset{\Pi_{3\times 4}}{\Longrightarrow} \left[\begin{array}{cccc} \boxed{2} & -3 & 0 & 3 \\ 4 & 5 & 1 & 7 \\ 2 & -1 & -3 & 5 \end{array}\right] \overset{\substack{r_2=r_2-\lambda r_1 \\ \lambda=\frac{4}{2}=2}}{\Longrightarrow} \left[\begin{array}{cccc} \boxed{2} & -3 & 0 & 3 \\ \underline{0} & 11 & 1 & 1 \\ 2 & -1 & -3 & 5 \end{array}\right]$

$\overset{\substack{r_3=r_3-\lambda r_1 \\ \lambda=\frac{2}{2}=1}}{\Longrightarrow} \left[\begin{array}{cccc} 2 & -3 & 0 & 3 \\ \underline{0} & \boxed{11} & 1 & 1 \\ \underline{0} & 2 & -3 & 2 \end{array}\right] \overset{\substack{r_3=r_3-\lambda r_2 \\ \lambda=2/11}}{\Longrightarrow} \left[\begin{array}{cccc} 2 & -3 & 0 & 3 \\ \underline{0} & 11 & 1 & 1 \\ \underline{0} & \underline{0} & \boxed{-\frac{35}{11}} & \frac{20}{11} \end{array}\right] \Longrightarrow \left[\begin{array}{ccc|c} 2 & -3 & 0 & 3 \\ 0 & 11 & 1 & 1 \\ 0 & 0 & -35/11 & 20/11 \end{array}\right]$

$\Longrightarrow \left\{\begin{array}{l} 2u-3v+0w=3 \\ 0u+11v+1w=1 \\ 0u-0v-\frac{35}{11}w=\frac{20}{11} \end{array}\right. \overset{r_3:}{\Longrightarrow} \left\{\begin{array}{l} 2u-3v+0w=3 \\ 0u+11v+1w=1 \\ -\frac{35}{11}w=\frac{20}{11} \end{array}\right. \Longrightarrow \left\{\begin{array}{l} 2u-3v+0w=3 \\ 0u+11v+1w=1 \\ w=-\frac{4}{7} \end{array}\right. \Longrightarrow$

$\overset{r_2:}{\Longrightarrow} \left\{\begin{array}{l} 2u-3v+0w=3 \\ 11v+1\cdot(-\frac{4}{7})=1 \\ w=-\frac{4}{7} \end{array}\right. \Longrightarrow \left\{\begin{array}{l} 2u-3v+0w=3 \\ v=\frac{1}{7} \\ w=-\frac{4}{7} \end{array}\right. \overset{r_1:}{\Longrightarrow} \left\{\begin{array}{l} 2u-3\cdot(\frac{1}{7})=3 \\ v=\frac{1}{7} \\ w=-\frac{4}{7} \end{array}\right. \Longrightarrow$

$\overset{\cdots}{\Longrightarrow} \left\{\boxed{\begin{array}{l} u=12/7 \\ v=1/7 \\ w=-4/7 \end{array}}\right.$

· **2η Άσκηση:**

i) **Έχουμε:** $\left[\begin{array}{ccc} 4 & 0 & 1 \\ 0 & 1 & 0 \\ 4 & 0 & 0 \end{array}\right]_{3\times 3}^{A} \cdot \left[\begin{array}{c} 3 \\ 4 \\ 5 \end{array}\right]_{3\times 1}^{B} \overset{*}{=} \left[\begin{array}{c} \left[\begin{array}{ccc} 4 & 0 & 1 \end{array}\right]\cdot\left[\begin{array}{c} 3 \\ 4 \\ 5 \end{array}\right] \\ \left[\begin{array}{ccc} 0 & 1 & 0 \end{array}\right]\cdot\left[\begin{array}{c} 3 \\ 4 \\ 5 \end{array}\right] \\ \left[\begin{array}{ccc} 4 & 0 & 0 \end{array}\right]\cdot\left[\begin{array}{c} 3 \\ 4 \\ 5 \end{array}\right] \end{array}\right] =$

$= \left[\begin{array}{c} (4)(3)+(0)(4)+(1)(5) \\ (0)(3)+(1)(4)+(0)(5) \\ (4)(3)+(0)(4)+(0)(5) \end{array}\right] = \boxed{\left[\begin{array}{c} 21 \\ 4 \\ 12 \end{array}\right]_{3\times 1}^{C}}$

ii) **Έχουμε:** $\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]_{3\times 3}^{I} \cdot \left[\begin{array}{c} 5 \\ -2 \\ 3 \end{array}\right]_{3\times 1}^{D} \overset{*}{=} \left[\begin{array}{c} \left[\begin{array}{ccc} 1 & 0 & 0 \end{array}\right] \cdot \left[\begin{array}{c} 5 \\ -2 \\ 3 \end{array}\right] \\ \left[\begin{array}{ccc} 0 & 1 & 0 \end{array}\right] \cdot \left[\begin{array}{c} 5 \\ -2 \\ 3 \end{array}\right] \\ \left[\begin{array}{ccc} 0 & 0 & 1 \end{array}\right] \cdot \left[\begin{array}{c} 5 \\ -2 \\ 3 \end{array}\right] \end{array}\right] =$

$= \left[\begin{array}{c} (1)(5)+(0)(-2)+(0)(3) \\ (0)(5)+(1)(-2)+(0)(3) \\ (0)(5)+(0)(-2)+(1)(3) \end{array}\right] = \boxed{\left[\begin{array}{c} 5 \\ -2 \\ 3 \end{array}\right]_{3\times 1}^{E}}$

iii) **Έχουμε:** $\left[\begin{array}{cc} 2 & 0 \\ 1 & 3 \end{array}\right]_{2\times 2}^{F} \cdot \left[\begin{array}{c} 1 \\ 1 \end{array}\right]_{2\times 1}^{G} \overset{*}{=} \left[\begin{array}{c} \left[\begin{array}{cc} 2 & 0 \end{array}\right] \cdot \left[\begin{array}{c} 1 \\ 1 \end{array}\right] \\ \left[\begin{array}{cc} 1 & 3 \end{array}\right] \cdot \left[\begin{array}{c} 1 \\ 1 \end{array}\right] \end{array}\right] =$

$= \left[\begin{array}{c} (2)(1)+(0)(1) \\ (1)(1)+(1)(3) \end{array}\right] = \boxed{\left[\begin{array}{c} 2 \\ 4 \end{array}\right]_{2\times 1}^{H}}$

· **3η Άσκηση:**

Έχουμε: $\left\{\begin{array}{l} x_1-2x_2-3x_3=-1 \\ 3x_1+x_2+x_3=4 \\ 11x_1-x_2-3x_3=10 \end{array}\right. \overset{\Pi_{3\times 4}}{\Longrightarrow} \left[\begin{array}{cccc} \boxed{1} & -2 & -3 & -1 \\ 3 & 1 & 1 & 4 \\ 11 & -1 & -3 & 10 \end{array}\right] \overset{\substack{r_2=r_2-\lambda r_1 \\ \lambda=\frac{3}{1}=3}}{\Longrightarrow} \left[\begin{array}{cccc} \boxed{1} & -2 & -3 & -1 \\ \underline{0} & 7 & 10 & 7 \\ 11 & -1 & -3 & 10 \end{array}\right]$

$\overset{\substack{r_3=r_3-\lambda r_1 \\ \lambda=\frac{11}{1}=11}}{\Longrightarrow} \left[\begin{array}{cccc} 1 & -2 & -3 & -1 \\ \underline{0} & \boxed{7} & 10 & 7 \\ \underline{0} & 21 & 30 & 21 \end{array}\right] \overset{\substack{r_3=r_3-\lambda r_2 \\ \lambda=21/7}}{\Longrightarrow} \left[\begin{array}{cccc} 1 & -2 & -3 & -1 \\ \underline{0} & 7 & 10 & 7 \\ \underline{0} & \underline{0} & \boxed{0} & 0 \end{array}\right] \Longrightarrow \left[\begin{array}{cccc} 1 & -2 & -3 & -1 \\ 0 & 7 & 10 & 7 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$\Longrightarrow \left\{\begin{array}{l} 1x_1-2x_2-3x_3=-1 \\ 0x_1+7x_2+10x_3=7 \\ 0x_1 \pm 0x_2 \pm 0x_3=0 \end{array}\right. \Longrightarrow \left\{\begin{array}{l} x_1-2x_2-3x_3=-1 \\ 7x_2+10x_3=7 \\ 0=0 \end{array}\right.$

(α) Άπειρες λύσεις (αόριστο σύστημα) ...

(β) Ένα σύστημα ονομάζεται ιδιόμορφο (singular) όταν ο πίνακας συντελεστών του έχει ορίζουσα που ισούτε με μηδέν. Αυτό σημαίνει ότι οι γραμμές ή οι στήλες του πίνακα είναι γραμμικά εξαρτημένες, και το σύστημα είτε δεν έχει λύση είτε έχει άπειρες λύσεις.

Για παράδειγμα: x+y=6 **και** 2x+2y=12 .

· **5η Άσκηση:**

(α) **Έχουμε:** $\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 5 & 7 & 8 \end{array}\right]^{A}_{3\times 3} \overset{*}{\Longrightarrow} \left[\begin{array}{ccc|ccc} \boxed{1} & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 5 & 7 & 8 & 0 & 0 & 1 \end{array}\right]^{A|I}_{3\times 6} \Longrightarrow$

$\overset{r_2=r_2-\lambda r_1}{\underset{\lambda=1/1=1}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} \boxed{1} & 1 & 1 & 1 & 0 & 0 \\ \underline{0} & 1 & 1 & -1 & 1 & 0 \\ 5 & 7 & 8 & 0 & 0 & 1 \end{array}\right] \overset{r_3=r_3-\lambda r_1}{\underset{\lambda=5/1=5}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ \underline{0} & \boxed{1} & 1 & -1 & 1 & 0 \\ \underline{0} & 2 & 3 & -5 & 0 & 1 \end{array}\right] \overset{r_3=r_3-\lambda r_2}{\underset{\lambda=2/1=2}{\Longrightarrow}}$

$\overset{r_3=r_3-\lambda r_2}{\underset{\lambda=2/1=2}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ \underline{0} & 1 & 1 & -1 & 1 & 0 \\ \underline{0} & \underline{0} & \boxed{1} & -3 & -2 & 1 \end{array}\right] \Longrightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -3 & -2 & 1 \end{array}\right] \Longrightarrow$

$\overset{r_2=r_2-\lambda r_3}{\underset{\lambda=1}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & \underline{0} & 2 & 3 & -1 \\ 0 & 0 & 1 & -3 & -2 & 1 \end{array}\right] \overset{r_1=r_1-\lambda r_3}{\underset{\lambda=1}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 1 & 1 & \underline{0} & 4 & 2 & -1 \\ 0 & 1 & \underline{0} & 2 & 3 & -1 \\ 0 & 0 & 1 & -3 & -2 & 1 \end{array}\right] \overset{r_1=r_1-\lambda r_2}{\underset{\lambda=1}{\Longrightarrow}}$

$\overset{r_1=r_1-\lambda r_2}{\underset{\lambda=1}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 1 & \underline{0} & \underline{0} & 2 & -1 & 0 \\ 0 & 1 & \underline{0} & 2 & 3 & -1 \\ 0 & 0 & 1 & -3 & -2 & 1 \end{array}\right]^{I|A^{-1}}_{3\times 6} \overset{\cdots}{\Longrightarrow} \boxed{\left[\begin{array}{ccc} 3 & -1 & 0 \\ 1 & 3 & -1 \\ -3 & -2 & 1 \end{array}\right]^{A^{-1}}_{3\times 3}}$

(β) **Έχουμε:** $\left[\begin{array}{ccc} 6 & -2 & 1 \\ -4 & 1 & -1 \\ 1 & 0 & 1 \end{array}\right]^{B}_{3\times 3} \overset{*}{\Longrightarrow} \left[\begin{array}{ccc|ccc} \boxed{6} & -2 & 1 & 1 & 0 & 0 \\ -4 & 1 & -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right]^{B|I}_{3\times 6} \Longrightarrow$

$\overset{r_2=r_2-\lambda r_1}{\underset{\lambda=-4/6=-2/3}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} \boxed{6} & -2 & 1 & 1 & 0 & 0 \\ \underline{0} & -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \overset{r_3=r_3-\lambda r_1}{\underset{\lambda=1/6}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 6 & -2 & 1 & 1 & 0 & 0 \\ \underline{0} & \boxed{-\frac{1}{3}} & -\frac{1}{3} & \frac{2}{3} & 1 & 0 \\ \underline{0} & -\frac{1}{3} & \frac{5}{6} & -\frac{1}{6} & 0 & 1 \end{array}\right] \overset{r_3=r_3-\lambda r_2}{\underset{\lambda=1\ldots}{\Longrightarrow}}$

$\overset{r_3=r_3-\lambda r_2}{\underset{\lambda=1\ldots}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 6 & -2 & 1 & 1 & 0 & 0 \\ \underline{0} & -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 1 & 0 \\ \underline{0} & \underline{0} & \boxed{\frac{7}{6}} & -\frac{5}{6} & -1 & 1 \end{array}\right] \Longrightarrow \left[\begin{array}{ccc|ccc} 6 & -2 & 1 & 1 & 0 & 0 \\ 0 & -1/3 & -1/3 & 2/3 & 1 & 0 \\ 0 & 0 & 7/6 & -5/6 & -1 & 1 \end{array}\right] \Longrightarrow$

$\overset{r_3=r_3\cdot\frac{6}{7}}{\Longrightarrow} \left[\begin{array}{ccc|ccc} 6 & -2 & 1 & 1 & 0 & 0 \\ 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 1 & 0 \\ 0 & 0 & \underline{1} & -\frac{5}{7} & -\frac{6}{7} & \frac{6}{7} \end{array}\right] \overset{r_2=r_2\cdot(-3)}{\Longrightarrow} \left[\begin{array}{ccc|ccc} 6 & -2 & 1 & 1 & 0 & 0 \\ 0 & \underline{1} & \underline{1} & -2 & -3 & 0 \\ 0 & 0 & \underline{1} & -\frac{5}{7} & -\frac{6}{7} & \frac{6}{7} \end{array}\right] \overset{r_2=r_2-\lambda r_3}{\underset{\lambda=1}{\Longrightarrow}}$

$\overset{r_2=r_2-\lambda r_3}{\underset{\lambda=1}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} 6 & -2 & 1 & 1 & 0 & 0 \\ 0 & \underline{1} & \underline{0} & -\frac{9}{7} & -\frac{15}{7} & -\frac{6}{7} \\ 0 & 0 & \underline{1} & -\frac{5}{7} & -\frac{6}{7} & \frac{6}{7} \end{array}\right] \overset{r_1=r_1\cdot\frac{1}{6}}{\Longrightarrow} \left[\begin{array}{ccc|ccc} \underline{\frac{1}{6}} & -\frac{1}{3} & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ 0 & \underline{1} & \underline{0} & -\frac{9}{7} & -\frac{15}{7} & -\frac{6}{7} \\ 0 & 0 & \underline{1} & -\frac{5}{7} & -\frac{6}{7} & \frac{6}{7} \end{array}\right] \overset{r_1=r_1-\lambda r_3}{\underset{\lambda=1/6}{\Longrightarrow}}$

$\overset{r_1=r_1-\lambda r_3}{\underset{\lambda=1/6}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} \underline{\frac{1}{6}} & -\frac{1}{3} & \underline{0} & \frac{2}{7} & \frac{1}{7} & -\frac{1}{7} \\ 0 & \underline{1} & \underline{0} & -\frac{9}{7} & -\frac{15}{7} & -\frac{6}{7} \\ 0 & 0 & \underline{1} & -\frac{5}{7} & -\frac{6}{7} & \frac{6}{7} \end{array}\right] \overset{r_1=r_1-\lambda r_2}{\underset{\lambda=-1/3}{\Longrightarrow}} \left[\begin{array}{ccc|ccc} \underline{\frac{1}{6}} & \underline{0} & \underline{0} & -\frac{1}{7} & -\frac{4}{7} & -\frac{6}{7} \\ 0 & \underline{1} & \underline{0} & -\frac{9}{7} & -\frac{15}{7} & -\frac{6}{7} \\ 0 & 0 & \underline{1} & -\frac{5}{7} & -\frac{6}{7} & \frac{6}{7} \end{array}\right]^{I\ |\ B^{-1}}_{3\times 6} \Longrightarrow$

$\overset{\cdots}{\Longrightarrow} \boxed{\left[\begin{array}{ccc} -\frac{1}{7} & -\frac{4}{7} & -\frac{6}{7} \\ -\frac{9}{7} & -\frac{15}{7} & -\frac{6}{7} \\ -\frac{5}{7} & -\frac{6}{7} & \frac{6}{7} \end{array}\right]^{B^{-1}}_{3\times 3}} \overset{\cdots}{\Longrightarrow} \left[\begin{array}{ccc} -1/7 & -4/7 & -6/7 \\ -9/7 & -15/7 & -6/7 \\ -5/7 & -6/7 & +6/7 \end{array}\right]^{B^{-1}}_{3\times 3}$

· **6η Άσκηση:**

Α. **Εάν το μοναδικό στοιχείο του είναι διάφορο του μηδενός.**

Β. **Μόνο όταν η ορίζουσα του είναι διάφορη του μηδενός.**

Γ. **Μόνο όταν όλα τα στοιχεία της διαγωνίου του είναι διάφορα του μηδενός.**

· **7η Άσκηση:**

(α) **Είναι:** $AB = AC \Rightarrow A^{-1}(AB) = A^{-1}(AC) \Rightarrow (A^{-1}A)B = (A^{-1}A)C \Rightarrow IB = IC \Rightarrow \boxed{B = C}$

(β) **Ένα π.χ. είναι:** $\boxed{B_{2\times 2} = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}}$ **και** $\boxed{C_{2\times 2} = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}}$

· **8η Άσκηση:**

Έχουμε: $A_{3\times 2} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 4 \end{bmatrix}$ **και** $B_{3\times 2} = \begin{bmatrix} 1 & 3 \\ 4 & 2 \\ 5 & 6 \end{bmatrix}$

Αρχικά: $A^{\tau}_{2\times 3} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 4 \end{bmatrix}$ **και** $B^{\tau}_{2\times 3} = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 6 \end{bmatrix}$ **(αλλαγή γραμμών με στήλλες ...)**

$\alpha)$ $A^{\tau}_{2\times 3} \cdot B_{3\times 2} = C^{1}_{2\times 2} = \begin{bmatrix} (2)(1)+(1)(4)+(3)(5) & (2)(3)+(1)(2)+(3)(6) \\ (1)(1)+(2)(4)+(4)(5) & (1)(3)+(2)(2)+(4)(6) \end{bmatrix} = \boxed{\begin{bmatrix} 21 & 26 \\ 29 & 31 \end{bmatrix}}$

$\beta)$ $B^{\tau}_{2\times 3} \cdot A_{3\times 2} = C^{2}_{2\times 2} = \begin{bmatrix} (1)(2)+(4)(1)+(5)(3) & (1)(1)+(4)(2)+(5)(4) \\ (3)(2)+(2)(1)+(6)(3) & (3)(1)+(2)(2)+(6)(4) \end{bmatrix} = \boxed{\begin{bmatrix} 21 & 29 \\ 26 & 31 \end{bmatrix}}$

$\gamma)$ $A_{3\times 2} \cdot B^{\tau}_{2\times 3} = C^{3}_{3\times 3} = \begin{bmatrix} (2)(1)+(1)(3) & (2)(4)+(1)(2) & (2)(5)+(1)(6) \\ (1)(1)+(2)(3) & (1)(4)+(2)(2) & (1)(5)+(2)(6) \\ (3)(1)+(4)(3) & (3)(4)+(4)(2) & (3)(5)+(4)(6) \end{bmatrix} = \boxed{\begin{bmatrix} 5 & 10 & 16 \\ 7 & 8 & 17 \\ 15 & 20 & 39 \end{bmatrix}}$

$\delta)$ $B_{3\times 2} \cdot A^{\tau}_{2\times 3} = C^{4}_{3\times 3} = \begin{bmatrix} (1)(2)+(3)(1) & (1)(1)+(3)(2) & (1)(3)+(3)(4) \\ (4)(2)+(2)(1) & (4)(1)+(2)(2) & (4)(3)+(2)(4) \\ (5)(2)+(6)(1) & (5)(1)+(6)(2) & (5)(3)+(6)(4) \end{bmatrix} = \boxed{\begin{bmatrix} 5 & 7 & 15 \\ 10 & 8 & 20 \\ 16 & 17 & 39 \end{bmatrix}}$

· **10η Άσκηση:**

$\alpha)$ **Έχουμε:** $A^1_{2\times 2} = \begin{bmatrix} \boxed{3} & 2 \\ 12 & 9 \end{bmatrix} \underset{\underline{\lambda=4^*}}{\overset{r_2=r_2-\lambda r_1}{\Longrightarrow}} \begin{bmatrix} 3 & 2 \\ 0 & \boxed{1} \end{bmatrix} \Longrightarrow \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}^*$

Άρα: $\boxed{U_{2\times 2} = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix} \quad \textbf{και} \quad L_{2\times 2} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}$

$\beta)$ **Έχουμε:** $A^2_{3\times 3} = \begin{bmatrix} \boxed{2} & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \underset{\underline{\lambda=1/2}}{\overset{r_2=r_2-\lambda r_1}{\Longrightarrow}} \begin{bmatrix} \boxed{2} & 1 & 1 \\ 0 & \frac{3}{2} & \frac{1}{2} \\ 1 & 1 & 2 \end{bmatrix} \underset{\underline{\lambda=1/2}}{\overset{r_3=r_3-\lambda r_1}{\Longrightarrow}} \begin{bmatrix} 2 & 1 & 1 \\ 0 & \boxed{\frac{3}{2}} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{3}{2} \end{bmatrix}$

$$\underset{\underline{\lambda=1/3}}{\overset{r_3=r_3-\lambda r_2}{\Longrightarrow}} \begin{bmatrix} 2 & 1 & 1 \\ 0 & \frac{3}{2} & \frac{1}{2} \\ 0 & 0 & \boxed{\frac{4}{3}} \end{bmatrix} \Longrightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3/2 & 1/2 \\ 0 & 0 & 4/3 \end{bmatrix}$$

Άρα: $\boxed{U_{3\times 3} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3/2 & 1/2 \\ 0 & 0 & 4/3 \end{bmatrix} \quad \textbf{και} \quad L_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 1/2 & 1/3 & 1 \end{bmatrix}}$

$\gamma)$ **Έχουμε:** $A^3_{3\times 3} = \begin{bmatrix} \boxed{1} & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 9 \end{bmatrix} \underset{\underline{\lambda=1}}{\overset{r_2=r_2-\lambda r_1}{\Longrightarrow}} \begin{bmatrix} \boxed{1} & 1 & 1 \\ 0 & 2 & 2 \\ 1 & 3 & 9 \end{bmatrix} \underset{\underline{\lambda=1}}{\overset{r_3=r_3-\lambda r_1}{\Longrightarrow}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & \boxed{2} & 2 \\ 0 & 2 & 8 \end{bmatrix}$

$$\underset{\underline{\lambda=1}}{\overset{r_3=r_3-\lambda r_2}{\Longrightarrow}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & \boxed{6} \end{bmatrix} \Longrightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 6 \end{bmatrix}$$

Άρα: $\boxed{U_{3\times 3} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 6 \end{bmatrix} \quad \textbf{και} \quad L_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}$

· **11η Άσκηση:**

$\alpha)$ **Έχουμε:** $A^{\alpha}_{3\times 3} = \begin{bmatrix} \boxed{0}^{*} & 1 & 1 \\ 1 & 0 & 1 \\ 4 & 5 & 6 \end{bmatrix} \overset{\substack{r_1\leftarrow r_3 \\ r_3\leftarrow r_1}}{\Longrightarrow} \begin{bmatrix} \boxed{4} & 5 & 6 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ **και έτσι** $P_{3\times 3} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$$\overset{\substack{r_2=r_2-\lambda r_1 \\ \Longrightarrow \\ \underline{\lambda=1/4}}}{} \begin{bmatrix} 4 & 5 & 6 \\ 0 & \boxed{-\frac{5}{4}} & -\frac{1}{2} \\ 0 & 1 & 1 \end{bmatrix} \overset{\substack{r_3=r_3-\lambda r_2 \\ \Longrightarrow \\ \underline{\lambda=-4/5}}}{} \begin{bmatrix} 4 & 5 & 6 \\ 0 & -\frac{5}{4} & -\frac{1}{2} \\ 0 & 0 & \boxed{\frac{3}{5}} \end{bmatrix} \overset{\cdots}{\Longrightarrow} \begin{bmatrix} 4 & 5 & 6 \\ 0 & -5/4 & -1/2 \\ 0 & 0 & 3/5 \end{bmatrix}$$

Άρα $L_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 1/4 & 1 & 0 \\ 0 & -4/5 & 1 \end{bmatrix}$ **και** $D_{3\times 3} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -5/4 & 0 \\ 0 & 0 & 3/5 \end{bmatrix}$ **και** $U_{3\times 3} = \begin{bmatrix} 1 & 5/4 & 3/2 \\ 0 & 1 & 2/5 \\ 0 & 0 & 1 \end{bmatrix}$

Τελικά:
$$\boxed{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}^{P} \cdot \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 4 & 5 & 6 \end{bmatrix}^{A} = \begin{bmatrix} 1 & 0 & 0 \\ 1/4 & 1 & 0 \\ 0 & -4/5 & 1 \end{bmatrix}^{L} \cdot \begin{bmatrix} 4 & 0 & 0 \\ 0 & -5/4 & 0 \\ 0 & 0 & 3/5 \end{bmatrix}^{D} \cdot \begin{bmatrix} 1 & 5/4 & 3/2 \\ 0 & 1 & 2/5 \\ 0 & 0 & 1 \end{bmatrix}^{U}}$$

$\beta)$ **Έχουμε:** $A^{\beta}_{3\times 3} = \begin{bmatrix} \boxed{3} & 6 & 3 \\ 6 & 12 & 6 \\ 3 & 3 & 3 \end{bmatrix} \overset{\substack{r_2=r_2-\lambda r_1 \\ \Longrightarrow \\ \underline{\lambda_{21}=2}}}{} \begin{bmatrix} \boxed{3} & 6 & 3 \\ 0 & 0 & 0 \\ 3 & 3 & 3 \end{bmatrix} \overset{\substack{r_3=r_3-\lambda r_1 \\ \Longrightarrow \\ \underline{\lambda_{31}=1}}}{} \begin{bmatrix} 3 & 6 & 3 \\ 0 & \boxed{0}^{*} & 0 \\ 0 & -3 & 0 \end{bmatrix}$

$$\overset{\substack{r_2\leftarrow r_3 \\ \Longrightarrow \\ r_3\leftarrow r_2}}{} \begin{bmatrix} 3 & 6 & 3 \\ 0 & \boxed{-3} & 0 \\ 0 & 0 & 0 \end{bmatrix} \Longrightarrow \begin{bmatrix} 3 & 6 & 3 \\ 0 & -3 & 0 \\ 0 & 0 & \boxed{0}^{!} \end{bmatrix} \overset{\substack{? \\ \Longrightarrow \\ \cdots}}{} \begin{bmatrix} 3 & 6 & 3 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix}^{***}$$

Αφού ο οδηγός είνια μηδέν και δεν γίνονται αντιμεταθέσεις, η παραγωντοποίηση δεν ολοκληρώνεται !

Τελικά:
$$\boxed{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}^{P} \cdot \begin{bmatrix} 3 & 6 & 3 \\ 6 & 12 & 6 \\ 3 & 3 & 3 \end{bmatrix}^{A} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{L} \cdot \begin{bmatrix} \\ \boxed{?} \\ \\ \end{bmatrix}^{D} \cdot \begin{bmatrix} 3 & 6 & 3 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix}^{U}}$$

*** Υπόθεση:** $D_{3\times 3} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0? \end{bmatrix}$

Ασκήσεις 4 και 9 είναι προγραμματιστικές ..!

ΤΕΛΟΣ